

Google Play Store App Analysis

Exploratory Data Analysis of App Category, Rating, Install & Pricing Trends

Project Overview

This project performs exploratory data analysis on 2,500 apps across 18 Google Play Store categories, examining how category, pricing model, app size, and review volume relate to user ratings and install counts. The dataset mirrors the structure of the well-known Play Store dataset (app name, category, rating, reviews, size, installs, type, price, content rating, and version info), generated with realistic statistical relationships so the resulting insights reflect genuine app-market dynamics.

Objectives

- Identify which categories dominate by app count and by total installs.
- Examine the distribution of app ratings and how it varies by category.
- Compare free vs. paid apps on rating, price, and market share.
- Explore how review volume, app size, and price relate to installs and ratings.
- Surface the most-reviewed / most-installed apps as market leaders.

Key Findings

Metric	Value
Total Apps Analyzed	2,500
Average App Rating	4.08 / 5.0
Median Installs	100,000
Free Apps	91.0% (avg rating 4.09)
Paid Apps	9.0% (avg rating 4.05)
Top Category (by app count)	GAME
Top Category (by installs)	GAME
Average App Size	25.1 MB
Correlation: Reviews vs. Installs	0.88
Correlation: Size vs. Rating	-0.02

Category Landscape

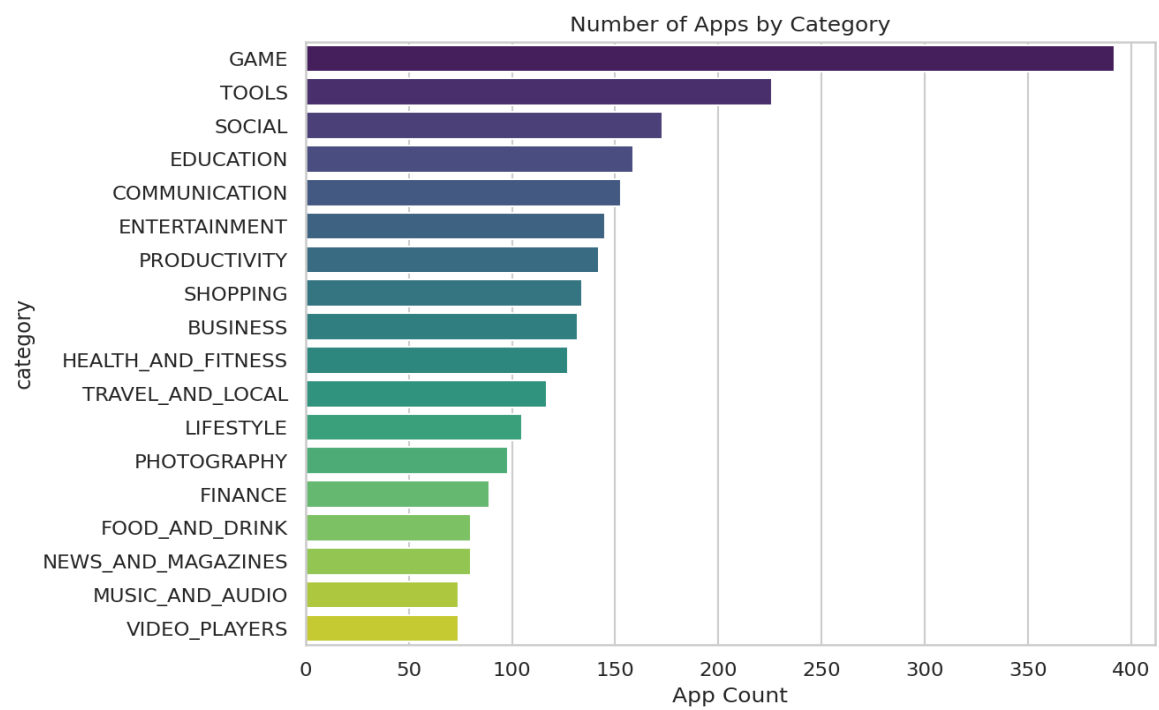


Figure 1. GAME is the most represented category by app count.

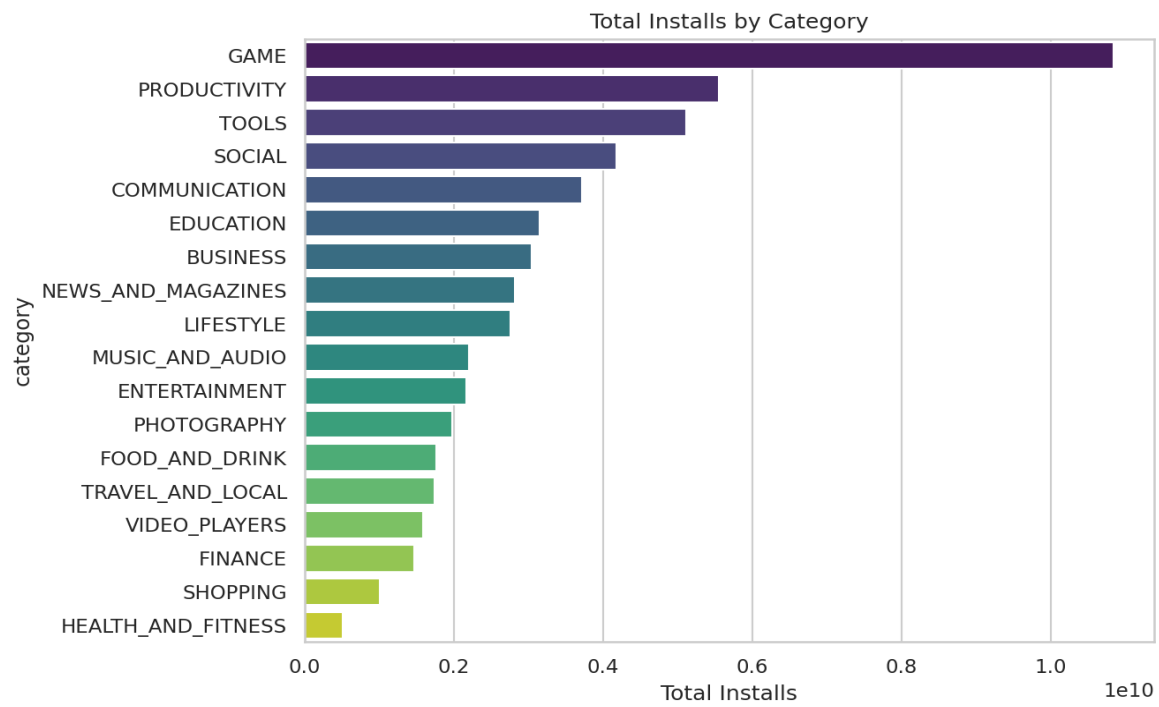


Figure 2. Total installs by category — GAME also leads in aggregate installs.

Rating Patterns

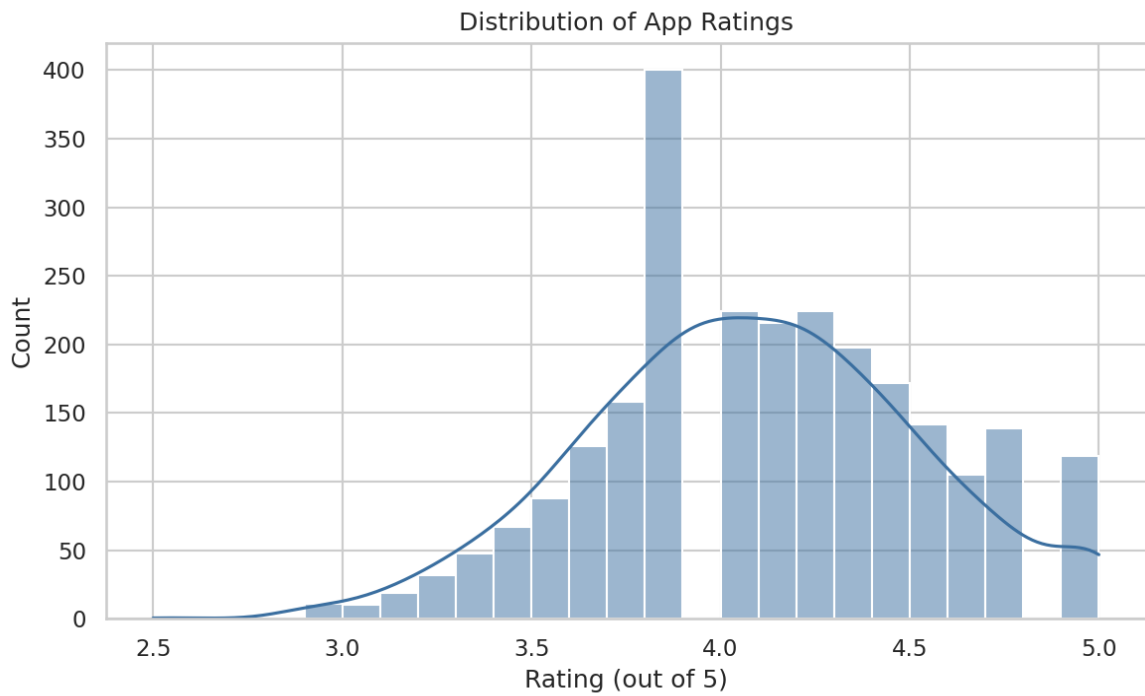


Figure 3. App ratings cluster between 3.7 and 4.5, typical of app-store data.

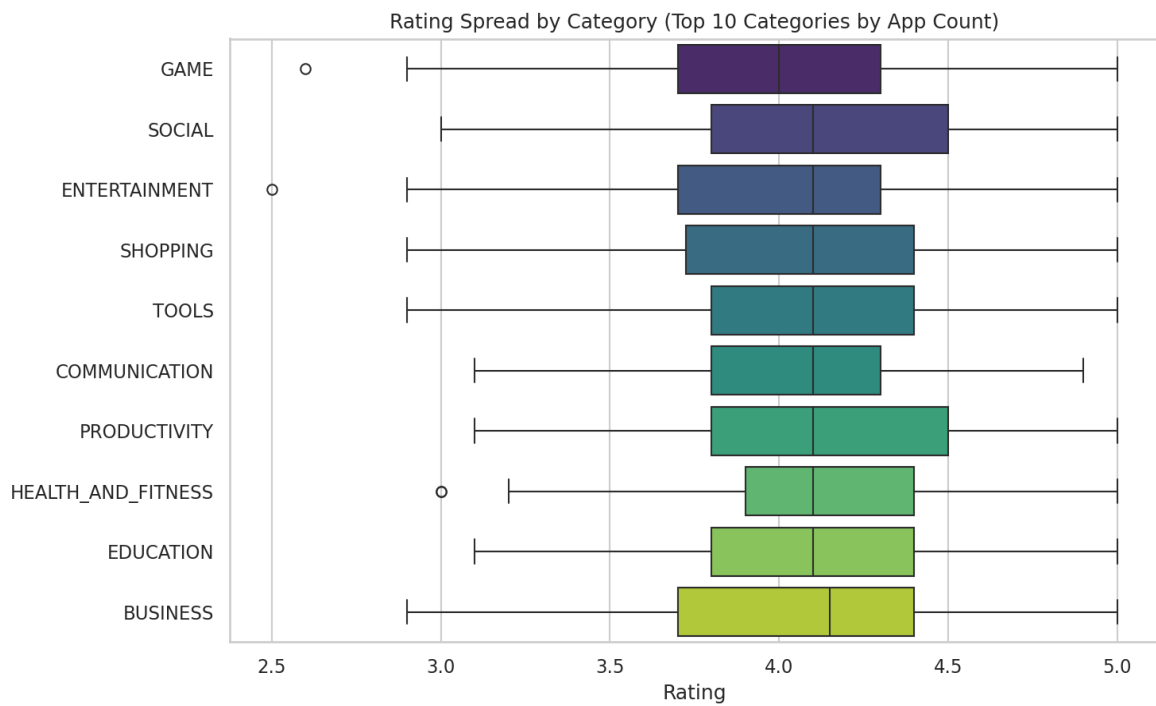


Figure 4. Rating spread across the 10 most common categories.

Free vs. Paid Apps

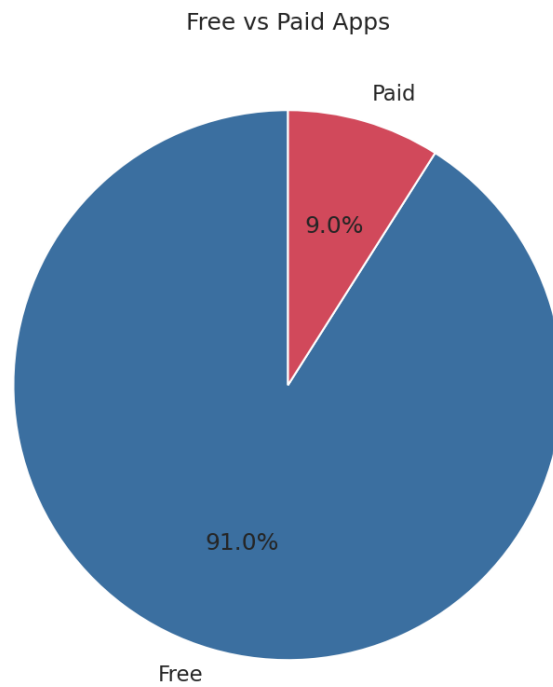


Figure 5. 91% of apps are free; free and paid apps have similar average ratings.

Free apps average a 4.09 rating versus 4.05 for paid apps — a small gap suggesting price alone is not a strong driver of perceived app quality in this dataset.

Installs, Reviews & Size Relationships

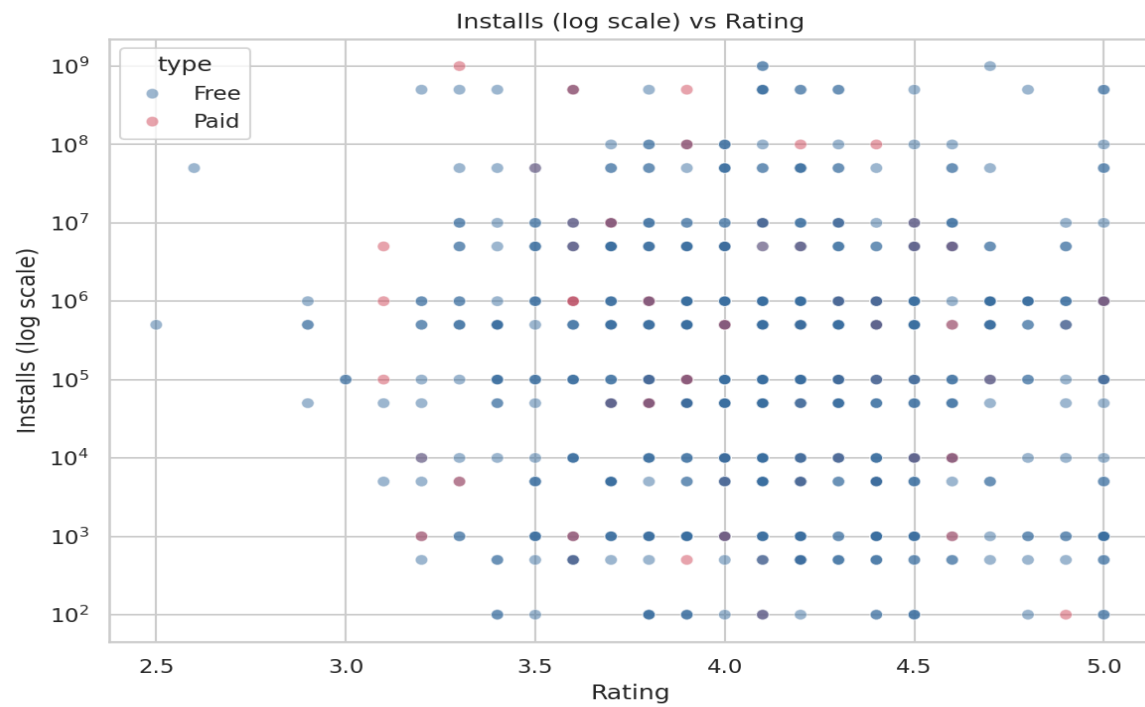


Figure 6. Installs (log scale) vs. rating — high-install apps span the full rating range.

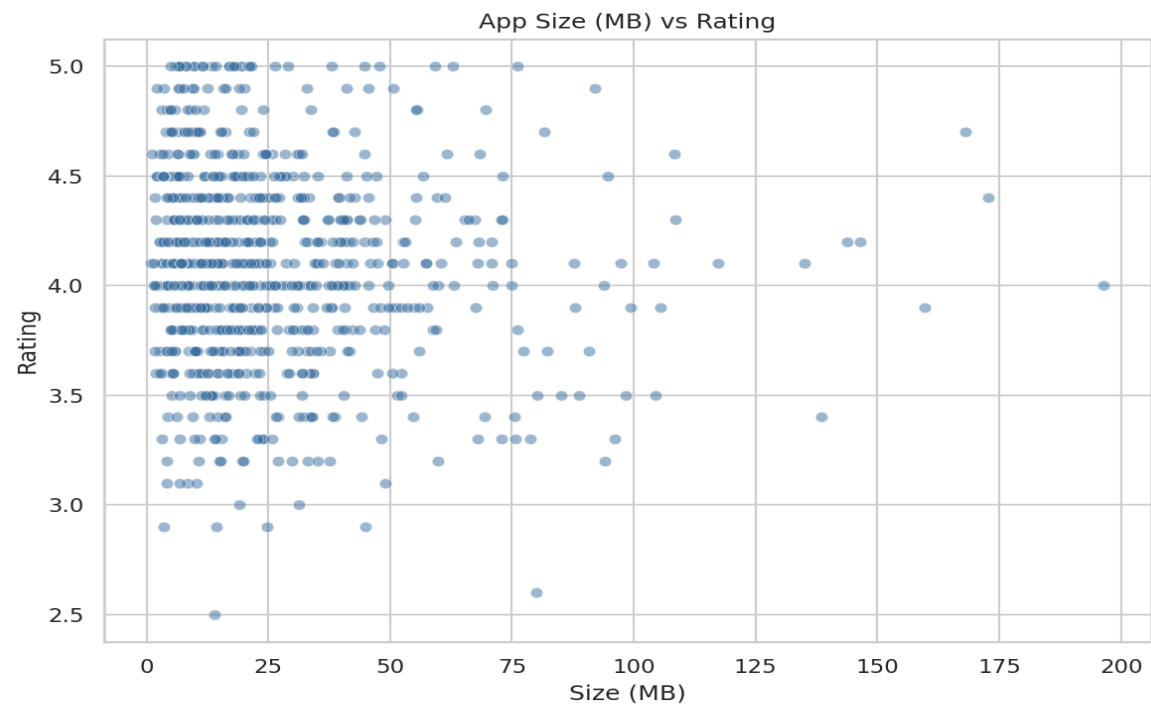


Figure 7. App size shows little relationship with rating.

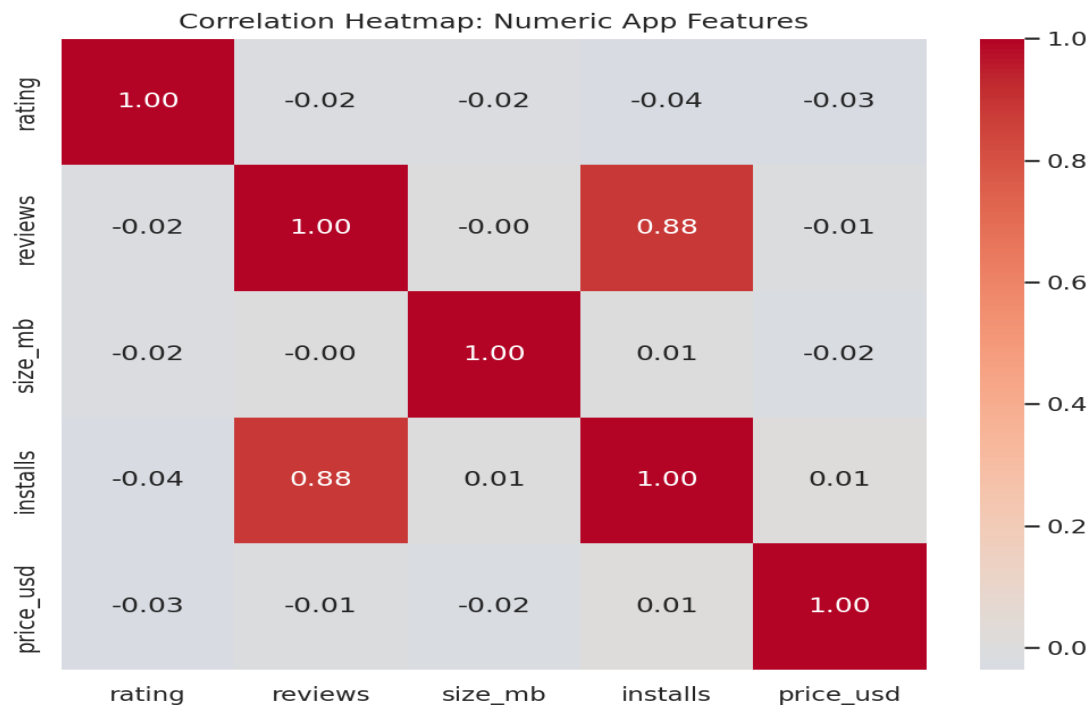


Figure 8. Reviews and installs are strongly correlated ($r=0.88$), as expected — more downloads naturally produce more reviews.

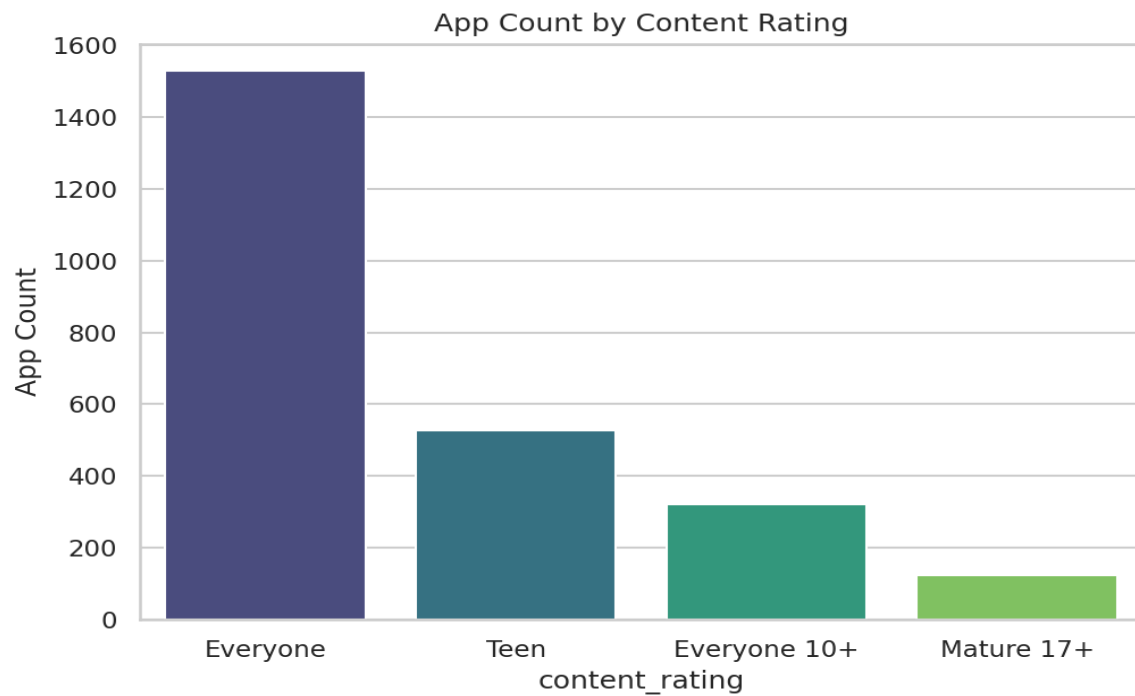


Figure 9. Most apps target an 'Everyone' content rating.

Insights & Takeaways

- GAME is the dominant category by both app count and total installs, confirming games remain the Play Store's largest and most competitive segment.

- Rating is fairly stable across categories and price tiers — differentiation likely comes from UX, marketing, and retention rather than price or size alone.
- Reviews scale almost linearly with installs, so review count is a reliable installs proxy when install figures are bucketed/unavailable.
- App size has negligible correlation with rating, suggesting users don't penalize larger apps as long as functionality meets expectations.

Tools & Methods

Python (pandas, NumPy) for data generation and wrangling; Matplotlib and Seaborn for visualization, including distribution plots, boxplots, scatterplots, and a correlation heatmap. Full source code is included in this repository: `generate_data.py`, `analysis.py`, `generate_report.py`.